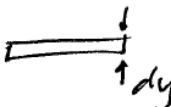


Product of Inertia by Integration
Example Problem Area Bounded by Two Lines

Product of Inertia by Integration

Definition: $I_{xy} = \int_A xy \, dA$ } Basically applies only to $dA = dx \, dy$ and double integration

If wish to use $dA =$ 

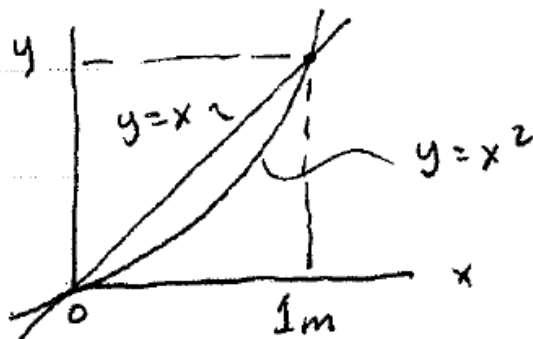
or $dA =$ , must use PAT in differential form for I_{xy} :

$$I_{xy} = \bar{I}_{x'y'} + A \cdot dx \, dy$$

$$dI_{xy} = d\bar{I}_{x'y'} + (dA) \tilde{x} \tilde{y}$$

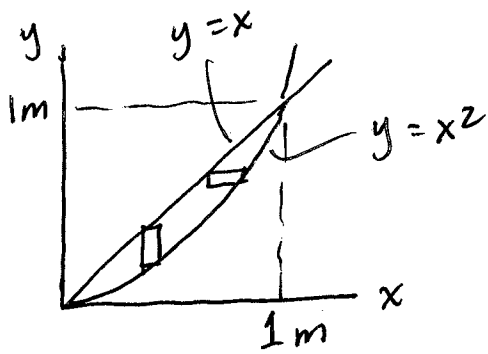
See
Hibbeler
Ex 10.6

Example Problem with Double Integral



$$I_{xy} = \int_A xy \, dA = \int_0^1 \int_{x^2}^x xy \, dy \, dx = \int_0^1 \frac{1}{2} (x^3 - x^5) \, dx = \frac{1}{2} \left(\frac{1}{4} - \frac{1}{6} \right) = \frac{1}{24} \, m^4 = I_{xy}$$

.0417



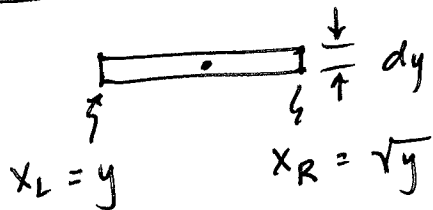
Use horiz element:

Use PAT in differential form:

$$dI_{xy} = \cancel{d\bar{I}_{xy}^0} + \tilde{x} \tilde{y} dA$$

$$I_{xy} = \int_A dI_{xy} = \int_A \tilde{x} \tilde{y} dA$$

Horiz Element



$$\begin{aligned} dA &= (x_R - x_L) dy \\ &= (\sqrt{y} - y) dy \end{aligned}$$

$$\tilde{y} = y$$

$$\tilde{x} = \frac{x_L + x_R}{2} = \left(\frac{y + \sqrt{y}}{2} \right)$$

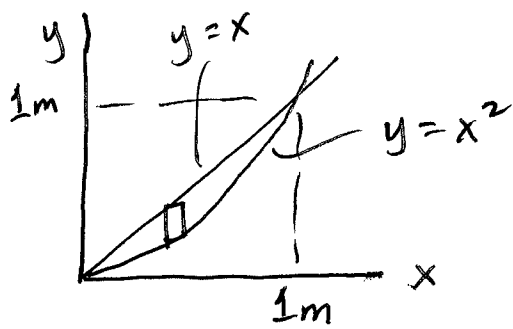
$$I_{xy} = \int_0^1 \underbrace{(y)}_{\tilde{y}} \underbrace{\left(\frac{y + \sqrt{y}}{2} \right)}_{\tilde{x}} \underbrace{(\sqrt{y} - y)}_{dA} dy$$

$$= \int_0^1 \frac{1}{2} y (-y^2 + y) dy = \int_0^1 \frac{1}{2} (y^2 - y^3) dy$$

$$= \frac{1}{2} \left(\frac{1}{3} y^3 - \frac{1}{4} y^4 \right) \Big|_0^1$$

$$= \frac{1}{2} \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{1}{2} \left(\frac{4}{12} - \frac{3}{12} \right) = \frac{1}{2} \left(\frac{1}{12} \right)$$

$$\boxed{I_{xy} = \frac{1}{24} \text{ m}^4 = 0.0417 \text{ m}^4}$$



Use Vertical Element

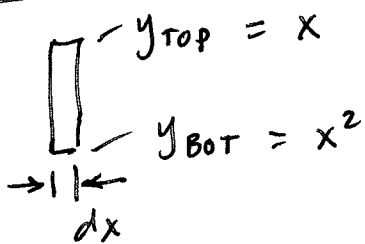
Use PAT in differential

form:

$$dI_{xy} = \cancel{d\bar{I}_{xy}} + \tilde{x} \tilde{y} dA$$

$$I_{xy} = \int_A dI_{xy} = \int_A \tilde{x} \tilde{y} dA$$

Vertical Element



$$dA = (y_{top} - y_{bot}) dx$$

$$= (x - x^2) dx$$

$$\tilde{x} = x$$

$$\tilde{y} = \frac{1}{2} (x^2 + x)$$

$$I_{xy} = \int_0^1 \underbrace{x}_{\tilde{x}} \underbrace{\left(\frac{1}{2} (x^2 + x) \right)}_{\tilde{y}} \underbrace{(x - x^2)}_{dA} dx$$

$$= \int_0^1 \frac{x}{2} (-x^4 + x^2) dx = \int_0^1 \frac{1}{2} (x^3 - x^5) dx$$

$$= \frac{1}{2} \left(\frac{1}{4} x^4 - \frac{1}{6} x^6 \right) \Big|_0^1 = \frac{1}{2} \left(\frac{1}{4} - \frac{1}{6} \right)$$

$$= \frac{1}{2} \left(\frac{3}{12} - \frac{2}{12} \right) = \frac{1}{2} \left(\frac{1}{12} \right)$$

$$\boxed{I_{xy} = \frac{1}{24} = .0417 \text{ m}^4}$$